

TrendPulse: Multi-Modal Social Media Trend Dynamics, Virality Prediction, and Conversational Topic Mining Framework

Course: B.Tech / BE in Artificial Intelligence and Machine Learning

Subject: Machine Learning Major Project

Case Study No.: 127 — Social Media Trend Analysis

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1. ABSTRACT

In the contemporary digital landscape, public social networks generate continuous, high-velocity streams of unstructured opinion, multimedia assets, and public discourse. Organizations seeking actionable intelligence face substantial challenges, including extreme power-law distributions in reach, severe noise-to-signal ratios, and non-stationary topical drift.

This major project develops TrendPulse, an end-to-end machine learning system that uncovers latent conversational themes, models engagement velocity, and predicts breakout virality from public social posts. Utilizing a curated multi-platform benchmark dataset of 5,000 public records across four ecosystems (Twitter/X, Instagram, LinkedIn, and Reddit) and six verticals, we construct a 278-dimensional feature space integrating natural language semantics (TF-IDF n-grams), lexical sentiment polarity and subjectivity, power-law follower scaling, and cyclical trigonometric temporal encodings.

We benchmark five supervised learning algorithms—Logistic Regression, Support Vector Machines (RBF Kernel), Random Forest, Gradient Boosting, and Multi-Layer Perceptron (MLP)—under 5-fold stratified cross-validation. Regularized Logistic Regression achieved the top overall performance with 57.10% test accuracy, 50.02% macro F1-score, and 72.82% multi-class ROC-AUC (OvR).

In parallel, unsupervised Latent Dirichlet Allocation (LDA) successfully segmented the corpus into six distinct conversational clusters. The complete framework is deployed as an interactive Streamlit dashboard providing real-time virality forecasting and automated content optimization heuristics.

2. PROBLEM DEFINITION & OBJECTIVES

2.1 Problem Statement (Case Study 127)

"An organization wants to identify meaningful patterns in publicly available social-media-related data (With Proper Justification)."

2.2 Mathematical Problem Formulation

Let each social media post i be represented by a multi-modal feature vector:

$$x_i = [x_{\text{text}}, x_{\text{sentiment}}, x_{\text{temporal}}, x_{\text{author}}, x_{\text{modality}}] \in \mathbb{R}^{278}$$

Where:

- $x_{\text{text}} \in \mathbb{R}^{250}$: Sublinear TF-IDF unigram and bigram document features.
- $x_{\text{sentiment}} \in [-1, 1] \times [0, 1]$: Emotional polarity and subjectivity scores.
- x_{temporal} : Cyclical temporal features represented using sine and cosine transformations.
- $x_{\text{author}} = [\log(1 + \text{followers}), \text{verified}]$: Logarithmically scaled author audience and verification status.
- $x_{\text{modality}} \in \{0, 1\}^4$: One-hot encoded media format (Text, Image, Video, Carousel).

The temporal features are represented as:

$$x_{\text{temporal}} = [\sin(2\pi h/24), \cos(2\pi h/24), \sin(2\pi d/7), \cos(2\pi d/7)]$$

where h represents the hour of the day and d represents the day of the week.

Target Formulation

The supervised classification task maps:

$$f: X \rightarrow Y$$

where:

$$Y = \{\text{Low}, \text{Moderate}, \text{Viral}\}$$

The classes are defined according to engagement quantiles:

- Low: Bottom 50%
- Moderate: 50%–85%
- Viral: Top 15% breakout engagement

2.3 Organizational Justification

PR Crisis Mitigation

Detecting emerging polarizing or negative themes before algorithmic amplification escalates potential brand damage.

Publishing Schedule Optimization

Identifying diurnal windows, particularly 12:00–14:00 and 18:00–21:00, that maximize reach per follower.

Content ROI Maximization

Evaluating whether expensive video production delivers a statistically verified engagement lift compared with static copy.

3. DATASET SPECIFICATION & PREPROCESSING

3.1 Dataset Schema

Volume: 5,000 public post records.

Platforms:

- Twitter/X — 25%
- Instagram — 25%
- LinkedIn — 25%
- Reddit — 25%

Verticals:

- Technology & AI
- Finance & Crypto
- Health & Wellness
- Entertainment & Culture
- Sports & Gaming
- Business & Careers

3.2 Preprocessing Pipeline

Text Normalization

Stripping HTTP URLs, user handles (@mentions), emojis, and non-informative punctuation; extracting hashtags as discrete tokens; and removing stopwords.

Power-Law Transformation

Applying:

$$\log(1+x)$$

to author follower counts to reduce the influence of extreme outlier accounts.

Cyclical Temporal Projection

Mapping 24-hour clocks and 7-day weeks to continuous sine and cosine coordinates, ensuring that Hour 23 and Hour 0 maintain true temporal proximity.

Categorical Encoding

One-Hot Encoding is applied to platform, domain, and media modality.

Stratification

A stratified 80/20 train-test split is used to preserve class ratios:

- Training: 4,000 samples
- Testing: 1,000 samples

4. EXPLORATORY DATA ANALYSIS (EDA) FINDINGS

4.1 Media Modality Lift

Video and Carousel formats achieve 2.1× to 2.4× higher median engagement than plain text across all platforms, potentially due to increased algorithmic dwell time.

4.2 Diurnal Peaks

Bimodal virality peaks occur during:

- 12:00–14:00 — Midday/lunch period
- 18:00–21:00 — Evening/leisure period

4.3 Subjectivity & Polarization Effect

Posts with high emotional subjectivity and polarized sentiment ($|p| > 0.30$) are 3.2× more likely to enter the Viral tier than neutral announcements.

4.4 Dialogue Hooks

Posts incorporating question marks generate 2.3× more discussion comments, which algorithms may weigh more heavily than passive likes.

5. EXPERIMENTAL RESULTS & MODEL BENCHMARKING

All models were evaluated using 5-fold stratified cross-validation on the 4,000 training samples and subsequently verified on the 1,000 held-out test samples.

Algorithm	Test Accuracy	Macro Precision	Macro Recall	Macro F1-Score	Weighted F1	Multi-Class ROC-AUC (OvR)	5-Fold CV F1 ($\mu \pm \sigma$)
Logistic Regression (Best)	57.10%	52.12%	49.29%	50.02%	55.72%	72.82%	50.10% ± 1.34%
Gradient Boosting	54.80%	49.69%	47.28%	48.00%	54.07%	72.27%	49.79% ± 1.23%
Random Forest	56.20%	53.94%	45.78%	46.42%	53.74%	71.95%	44.53% ± 1.05%

Support Vector Machine (SVM)	54.30%	49.08%	45.87 %	46.57 %	52.96%	71.70%	47.39% ± 1.05%
Multi-Layer Perceptron (MLP)	54.70%	48.14%	43.92 %	43.72 %	51.16%	71.69%	46.66% ± 1.66%

5.1 Analysis of Results

Why Linear Models Outperformed

In high-dimensional sparse text spaces ($D=278$), L2-regularized Logistic Regression builds a global convex decision boundary that can resist overfitting. Decision tree ensembles may split on sparse terms, potentially resulting in higher variance.

Top Virality Drivers

Gini importance indicates that the following features are among the dominant virality drivers:

- Emotional subjectivity: 0.135
- Follower reach: 0.115
- Video format: 0.066
- Posting hour: 0.097 combined

These features demonstrate the importance of emotional characteristics, audience size, content modality, and temporal timing in modeling social-media engagement.

6. UNSUPERVISED TOPIC DISCOVERY (LDA)

Latent Dirichlet Allocation (LDA) with $K=6$ autonomously discovered six thematic conversation clusters.

Topic 1 — Autonomous AI & Compute

Keywords: completely, open, championship, benchmark, weights, model

Topic 2 — Cinema & Visual Media

Keywords: day, office, record, looks, stunning, effects

Topic 3 — Macro Crypto & Markets

Keywords: bitcoin, investing, update, hyper, bull, economy

Topic 4 — Executive Strategy & Careers

Keywords: leadership, high, performance, futureofwork, productivity

Topic 5 — Global Esports & Sports

Keywords: records, breaks, million, viewers, world, championship

Topic 6 — Viral Culture & Tech Portfolios

Keywords: original, major, viral, truth, portfolio, problem

7. SYSTEM DEPLOYMENT & STREAMLIT WEB APP

The framework is deployed as an editorial Streamlit application featuring the following components:

7.1 Executive Problem Framing

Formal problem definitions, mathematical equations, and architecture flows.

7.2 Exploratory Data Analysis

Real-time cross-filtering, temporal heatmaps, format distributions, and sentiment boxplots.

7.3 Model Benchmark Arena

Side-by-side grouped bar charts, confusion matrices, and feature-importance rankings.

7.4 Real-Time Virality Predictor

A live testing console that accepts custom social-media copy and metadata to predict:

- Virality tier
- Virality index score (0–100)
- Automated editorial optimization heuristics

7.5 Topic Explorer

Interactive inspection of LDA vocabulary weights and discovered conversation clusters.

7.6 Viva Voce Defense Guide

Complete viva preparation materials and data-export facilities.

8. CONCLUSION & LIMITATIONS

8.1 Conclusion

Social-media virality can be modeled through multi-modal feature fusion and three-tier classification. The experimental results indicate that regularized linear decision boundaries can offer strong generalization in sparse text-feature settings compared with more complex ensemble and neural models.

The TrendPulse framework integrates supervised machine learning, unsupervised topic modeling, sentiment analysis, temporal feature engineering, and audience-related features into a unified system for social-media trend analysis.

8.2 Limitations

- Dark social sharing through encrypted messaging applications cannot be directly observed from public APIs.
- Follower-community graphs may require elevated API permissions.
- Social-media trends are highly dynamic and may change over time.

- Model performance is dependent on the quality and representativeness of the collected public dataset.
- Platform-specific algorithm changes can affect engagement patterns and reduce model stability over time.

8.3 Future Scope

Future improvements can include:

- Fine-tuning domain-specific Transformer embeddings such as RoBERTa and ModernBERT.
- Using Graph Neural Networks (GNNs) to model social diffusion cascades.
- Incorporating multimodal image and video embeddings.
- Performing real-time streaming analysis of social-media data.
- Developing platform-specific models to account for differences in recommendation and engagement mechanisms.
- Using explainable AI techniques to provide more transparent virality predictions.